Student's Index No.: _____ Signature: _____

UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: TEJS 232

COURSE TITLE: LEARNING, TEACHING AND APPLYING EUCLIDEAN GEOMETRY AND TRIGONOMETRY

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A

[25 marks]

Answer all the questions in this section.

1. If ΔABC and ΔXYZ are similar, and AB = 5 cm, BC = 8 cm, AC = 7 cm, and XY = 10 cm, what is the length of YZ?

   A. 11.2 cm

   B. 13.2 cm

   C. 16 cm

   D. 12.8 cm

2. Which of the following statements is always true about two similar triangles?

   A. Their areas are equal

   B. Their corresponding sides are equal in length

   C. Their corresponding angles are equal and sides proportional

   D. Their perimeters are equal

3. Which of the following set of numbers **cannot** form the sides of a right triangle?

A. (6, 8, 10)

B. (7, 24, 25)

C. (10, 26, 28)

D. (5, 12, 13)

4. If $\sin\theta = -\frac{\sqrt{3}}{2}$ and θ lies in the third quadrant, what is the value of tan θ?

A. $\sqrt{3}$

B. $-\sqrt{3}$

C. $\frac{1}{\sqrt{3}}$

D. $-\frac{1}{\sqrt{3}}$

5. Given that $\tan\theta = \frac{3}{4}$ and θ lies in the first quadrant, what is sin 2θ?

A. $\frac{24}{25}$

B. $\frac{7}{25}$

C. $\frac{3}{5}$

D. $\frac{4}{5}$

6. In triangle PQR, sides are p = 13 cm, q = 14 cm, and r = 15 cm. Which rule is most appropriate to find angle P?

A. Sine rule

B. Cosine rule

C. Pythagorean theorem

D. AA similarity

7. What is the general solution to the equation $\sin\theta = \frac{\sqrt{3}}{2}$ for θ between 0° and 360°?

A. 60°, 120°

B. 45°, 135°

C. 30°, 150°

D. 90°, 270°

8. In a circle, angle ABC is subtended by arc AC at the circumference, and angle AOC is subtended by the same arc at the center. If ∠ABC = 58°, what is ∠AOC?

A. 116°

B. 122°

C. 58°

D. 29°

9. In a cyclic quadrilateral, one of the interior angles is 78°. What is the measure of its opposite angle?

A. 102°

B. 78°

C. 180°

D. 108°

10. If $tan\ A = \frac{3}{4}$ and $tan\ B = \frac{2}{3}$, what is $tan\ (A + B)$?

A. $\frac{17}{12}$

B. $\frac{18}{11}$

C. $\frac{3}{2}$

D. $\frac{11}{18}$

11. If $\sin x = \frac{5}{13}$ and $cos\ x = \frac{12}{13}$, what is $sin(2x)$?

A. $\frac{60}{169}$

B. $\frac{120}{169}$

C. $\frac{120}{144}$

D. $\frac{130}{144}$

12. What is the maximum value of the function $f(\theta) = 3\cos\theta + 4\sin\theta$?

A. 4

B. 3

C. 5

D. 7

13. Which of the following is equivalent to $sin(A + B)$?

A. sin A sin B + cos A cos B

B. sin A cos B + cos A sin B

C. sin A cos B − cos A sin B

D. cos A cos B − sin A sin B

14. If $cos\ x = \frac{5}{13}$ and x is in quadrant IV, what is the exact value of sin(2x)?

A. $-\frac{120}{169}$

B. $\frac{120}{169}$

C. $-\frac{5}{13}$

D. $\frac{24}{25}$

15. Which of the following identities is not always true?

A. $1 + tan^2x = sec^2x$

B. $1 + cot^2x = csc^2x$

C. $sin^2x + tan^2x = 1$

D. $sin^2x + cos^2x = 1$

16. Which of these is the correct expression for cos(2A)?

A. $2\ cos^2A + 1$

B. $cos^2A - sin^2A$

C. $1 - 2\ sin^2A$

D. Both B and C

17. If $f(x) = 3\ sin\ x + 4\ cos\ x$, what is the maximum value of $f(x)$?

A. 5

B. 7

C. 1

D. $\sqrt{13}$

18. What is the standard form of the equation of a circle with center (3, −2) and radius 5?

A. $(x + 3)^2 + (y - 2)^2 = 25$

B. $(x - 3)^2 + (y + 2)^2 = 25$

C. $(x - 3)^2 + (y - 2)^2 = 5$

D. $(x - 3)^2 + (y + 2)^2 = 5$

19. In triangle ABC, side $a = 7$, side $b = 9$, and angle A= 50°. What is angle B?

A. 64°

B. 83°

C. 56°

D. 45°

20. What is the general form of the equation $(x - 2)^2 + (y + 5)^2 = 49$?

A. $x^2 + y^2 - 4x + 10y + 25 = 0$

B. $x^2 + y^2 - 4x - 10y - 20 = 0$

C. $x^2 + y^2 - 4x + 10y - 20 = 0$

D. $x^2 + y^2 + 4x - 10y = 49$

21. Given the equation: $x^2 + y^2 + 6x - 4y - 12 = 0$, what is the center and radius of the circle?

A. Center (−3, 2), Radius 5

B. Center (3, −2), Radius 5

C. Center (−3, 2), Radius $\sqrt{25}$

D. Center (−3, −2), Radius 5

22. Which situation requires the use of the Cosine Rule?

A. Two angles and a side

B. Two sides and an included angle

C. Two angles and the included side

D. One side and one angle

23. Which of the following equations represents a circle centered at the origin with radius **r**?

A. $x^2 + y^2 = r$

B. $(x + r)^2 + (y - r)^2 = 0$

C. $x^2 + y^2 = r^2$

D. $x^2 - y^2 = r^2$

24. If a circle has equation $x^2 + y^2 + 10x - 6y + 18 = 0$, what is the radius?

A. 4

B. 5

C. $\sqrt{7}$

D. 6

25. Given: $(x + 2)^2 + (y - 3)^2 = r^2$ passes through point (1, 1). What is the radius?

A. $\sqrt{13}$

B. 4

C. $\sqrt{10}$

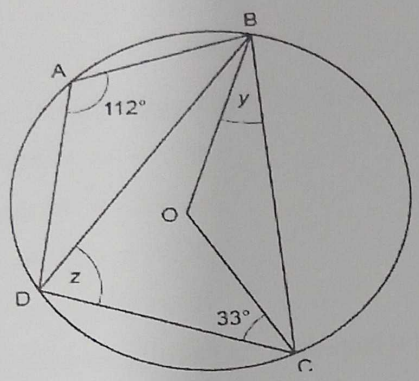
D. $\sqrt{20}$

SECTION B

[75 marks]

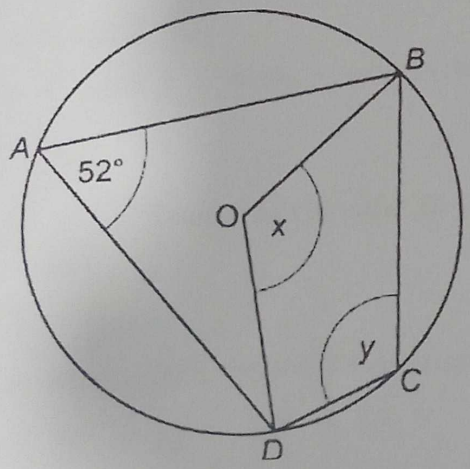
Answer three questions from this section.

1. A, B, C and D are points on the circumference of a circle, centre O. $\angle BAD = 112°$ and $\angle DCO = 33°$.
   a. Show $that\ y = 35°$. Give reasons for each stage of your working. ***(14 marks)***
   b. Work out angle z giving reasons for your answer. **(11 marks)**



Not to scale

2. A, B, C and D are points on the circumference of a circle, centre O.
   a. Work out $\angle BOD$. Give a reason for your answer. ***(6 marks)***
   b. Determine $\angle BCD$. Give a reason for your answer. ***(5 marks)***



Not to scale

   c. Find the exact value of $tan(75°)$ using the identity for $tan(A + B)$. ***(14 marks)***

3. With the aid of diagrams, explain **FIVE** (5) Euclidean Postulates. **(25 marks)**

4. a. Prove that: $1 + tan^2\theta = sec^2\theta$. ***(8 marks)***

b. Given the general form of a circle: $x^2 + y^2 - 6x + 4y + -12 = 0$,

i. Convert it into the standard form of a circle equation by completing the square. *(12 marks)*

ii. State the centre and radius of the circle. *(5 marks)*

5. Examine **FIVE** (5) misconceptions of teaching Euclidean & Non-Euclidean geometry and suggest ways to overcome each of them. *(25 marks)*